Super-resolution of Sentinel-2 satellite imagery

Reconstructing 5 m detail from 10 m imagery while keeping the band values that downstream analytics depend on

PROBLEM

Sentinel-2 imagery is free at 10 m per pixel. Commercial imagery at 1–2 m is not. We train a CNN to reconstruct fine spatial detail from the open imagery.

That output then feeds analytics. A prediction that looks sharper while shifting reflectance corrupts every index computed from it.

WHY THAT CONSTRAINT BITES

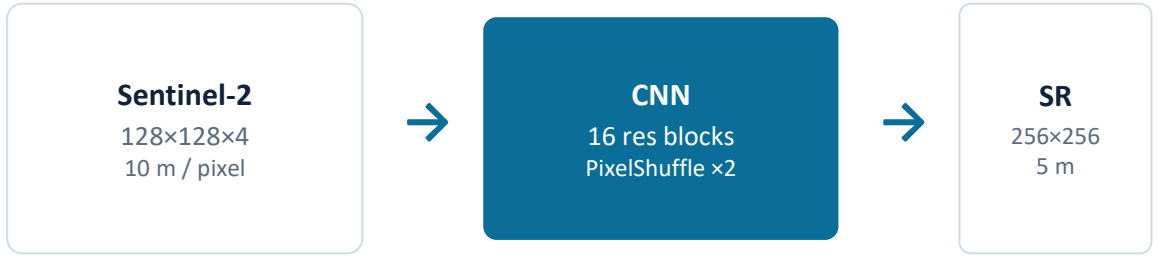
NDVI = (NIR − Red) / (NIR + Red)

Nudge NIR up 3% and Red down 2% and nobody sees a difference in the picture. NDVI moves, and every crop estimate downstream moves with it. Agriculture ministries, crop insurers and drought monitoring all run on this ratio.

WHAT WE ADD TO A STANDARD SR PIPELINE

- An observation constraint comparing the output with the INPUT, which the reconstruction loss never sees
- A confidence head, so the model reports where it should not be trusted
- A correctly specified π/2-cutoff decimation operator

INPUT AND OUTPUT



- Bands B2 blue, B3 green, B4 red, B8 near-infrared · all 10 m native
- Scale factor ×2 · 1.37 M parameters (+9.2 k for the confidence head)
- Reflectance, int16 scaled by 10000, clipped to [0, 1]
- Every metric computed on reflectance − never on an 8-bit display image

WHAT THE NETWORK IS ACTUALLY DOING

The network cannot recover information the sensor never captured. It makes an informed guess from the patterns it has seen. A model that guesses can also guess confidently and be wrong, which is the standard objection to super-resolution in operational remote sensing, and the reason we added a confidence map.

Data

SEN2VENµS: real captures at both resolutions, split by site so no test patch neighbours a training patch

THE DATASET

Sentinel-2 at 10 m paired with VENµS at 5 m: same place, same date, co-registered by the dataset authors.

Both sides are genuine captures. The lazy alternative is to blur and shrink sharp images to make your own low-resolution inputs. The network then learns to invert your blur, which is not the blur a real sensor produces: it scores beautifully and fails on real data.

Zenodo 14603764 · 29 sites · non-commercial licence

SPLIT PROTOCOL

Splits are by SITE, never by patch.

Patches inside one site are neighbouring crops of the same fields. A random patch split puts near-duplicates on both sides and the score stops meaning anything.

In-domain = held-out 10% of the training sites. It also selects the early-stopping checkpoint, so it is not a clean test set and is never labelled as one.

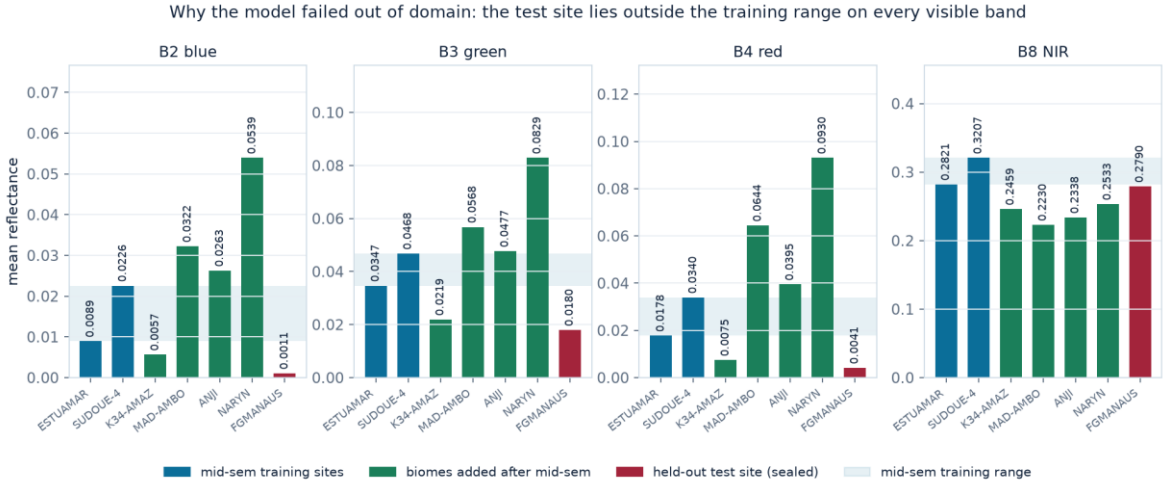
Out-of-domain = FGMANAUS, an entire site, sealed until the final score.

AUGMENTATION

Flips and 90-degree rotations (no 'up' in a top-down view), plus mixup and cutmix applied identically at both resolutions.

Added after the review: a per-band gain drawn log-uniformly and applied to the LOW and HIGH resolution patch ALIKE. Scaling both sides leaves the pair exactly as physically consistent as it was, while removing the absolute brightness of the training distribution as something the model can lean on.

SITES AND PER-BAND REFLECTANCE



Site (patches)	B4 red	B8 NIR	NDVI	Role
ESTUAMAR (911)	0.0178	0.2821	0.864	**train (mid-sem)**
SUDOUE-4 (935)	0.0340	0.3207	0.801	**train (mid-sem)**
K34-AMAZ (1384)	0.0075	0.2459	0.944	**train (added)**
MAD-AMBO (1442)	0.0644	0.2230	0.547	**train (added)**
ANJI (1500)	0.0395	0.2338	0.696	**train (added)**
NARYN (1500)	0.0930	0.2533	0.464	**train (added)**
FGMANAUS (129)	**0.0041**	0.2790	0.974	**TEST, sealed**

Adding biomes does not absorb the test site.

FGMANAUS red is 0.0041, still below the darkest training site. Widening the training range is not the same as swallowing the test site, and checking that is what keeps the comparison honest.

Network architecture · the backbone

EDSR-style residual CNN, 1.37 M parameters, upsampling once at the very end

FORWARD PATH FOR ONE PATCH

input
4×64×64

→

head conv
3×3, 4→64

→

body: 16 residual blocks
64 channels, at LOW resolution

→

tail
conv → PixelShuffle×2 → conv

→

SR output
4×128×128

long skip: body learns only what to ADD

RESIDUAL BLOCK

$$out = in + 0.1 \cdot [\ conv \rightarrow ReLU \rightarrow conv\ ](in)$$

The 0.1 is res_scale. Without it, 16 blocks at 64 channels diverge early in training. The addition is what lets gradients reach the first layers, so a deep stack stays trainable.

PARAMETERS

Component	Params	Share
head conv	2,368	0.2%
body, 16 residual blocks	1,216,512	88.7%
tail (conv, PixelShuffle, conv)	152,132	11.1%
confidence head (R4 only)	**+9,232**	**+0.7%**

IS THIS BACKBONE THE RIGHT ONE?

Architecture	Params	Held-out PSNR
SRNet, 16x64	1.37 M	**47.609 ±0.037**
SRCNN (2014)	75 k	pending
RCAN-lite, attention	1.53 M	pending
EDSR-large, 32x128	10.2 M	pending

DESIGN CHOICES, AND WHY

No BatchNorm anywhere

Normalising feature statistics throws away absolute intensity. For us that intensity IS the signal, since the whole point is keeping reflectance intact. EDSR dropped it for the same reason.

Upsample once, at the end

All 16 blocks run at 64×64 instead of 128×128, so the deep part costs a quarter as much.

PixelShuffle, not ConvTranspose

Transposed convolution leaves checkerboard artifacts. PixelShuffle produces 4 extra channels and interleaves them into a 2× bigger grid, so there is no overlap to go uneven.

L1, not L2

Squared error punishes large errors so hard that the network hedges and outputs blurry averages. L1 gives visibly sharper reconstructions.

Network architecture · heads, losses, and D

Each run changes exactly one component, so every comparison is an ablation rather than two unrelated systems

THE CONFIDENCE HEAD



Laplace negative log-likelihood

$$-\log p = |y - \mu| / b + \log 2b$$

Why Laplace and not Gaussian: training with L1 IS maximum likelihood under a Laplace model with a FIXED b. Letting the network predict b per pixel is the same objective with one assumption relaxed, so R4 differs from R1 by exactly that. A Gaussian head would generalise L2, the loss we already rejected for being blurry.

The pathology, and the fix

The NLL gradient on μ carries a factor 1/b, so pixels the model calls uncertain get a smaller mean gradient: hard pixels get labelled uncertain, stop being learned, and stay hard. We use β-NLL (Seitzer, ICLR 2022) at β = 1, making the mean gradient exactly sign(y − μ), identical to L1, while log b still converges to the local error magnitude.

The ablation did NOT confirm that. In-domain PSNR: β=0 43.84, β=0.5 43.85, β=1 43.68. Gradient clipping absorbs the inflation we guarded against, so β=0.5 is the better default. Our β=1 buys the lowest ENCE and is conservative on PSNR.

DEGRADATION OPERATOR D, USED BY M5 AND R3

x2 decimation keeps only frequencies below π/2, so D needs an anti-alias filter cut there. The 2×2 box used at mid-sem leaves 18% of surviving energy above π/2. Ten filters, all cut at π/2, are swept on slide 7.

THE RUNS: ONE CHANGE PER ROW

Run	Objective	Question it answers
R0	Bicubic ×2	The non-learned floor
R1	L1	Baseline network
R2	L1 + λ·SAM	Does a spectral term change anything?
R3	L1 + μ·|D(SR)−LR|	Does an observation constraint help?
R4	Laplace NLL + head	Is the predicted confidence calibrated?

Why R3 is different in kind

L1(SR, HR) compares the prediction with the TARGET and never looks at the input. A term built on the INPUT therefore cannot be implied by it. If the network invents detail that was never measured, shrinking the output back to 10 m will not reproduce the Sentinel-2 image, and this term notices. L1 cannot.

Iterative back-projection (Irani & Peleg 1991); the consistency property of Wald (1997).

Controls that make these ablations

- Identical architecture, schedule, step count, data and seeds; only the objective changes
- 3 seeds per row, reported mean ± population std. A difference inside the seed spread is not a difference
- The baseline gets the same effort as the new rows − quietly under-tuning R1 is the usual way a project manufactures an improvement

Results

Every learned row is 3 seeds, mean ± population std, scored on both regimes. Metrics on reflectance in [0, 1].

IN-DOMAIN · HELD-OUT PATCHES, TRAINING SITES

Picks the early-stopping checkpoint, so NOT a clean test set · and NOT comparable across the two blocks

Method	PSNR ↑	SSIM ↑	SAM ↓	NDVI MAE ↓	M5 ↓
R0 Bicubic ×2	42.159	0.9378	1.582	0.02810	0.00078
R1 CNN, L1	43.774 ±0.038	0.9506 ±0.0003	1.295 ±0.008	0.02415 ±0.00013	0.00464 ±0.00002
R2 CNN, L1 + λ·SAM	43.731 ±0.050	0.9500 ±0.0005	1.292 ±0.009	0.02413 ±0.00018	0.00460 ±0.00008
R3 CNN, L1 + μ·consist	43.717 ±0.051	0.9504 ±0.0005	1.301 ±0.010	0.02424 ±0.00018	0.00397 ±0.00006
R4 CNN, Laplace NLL	43.680 ±0.049	0.9498 ±0.0004	1.311 ±0.012	0.02438 ±0.00027	0.00461 ±0.00007
R1a R1 + augmentation	43.343 ±0.009	0.9476 ±0.0001	1.393 ±0.002	0.02524 ±0.00001	0.00389 ±0.00002
R1m R1, multi-biome	43.665 ±0.007	0.9561 ±0.0001	1.586 ±0.001	0.02741 ±0.00002	0.00278 ±0.00001
R1ma R1, multi-biome + aug	43.520 ±0.007	0.9551 ±0.0000	1.618 ±0.001	0.02783 ±0.00001	0.00241 ±0.00000
R4m R4, multi-biome + aug	43.475 ±0.003	0.9550 ±0.0000	1.629 ±0.000	0.02802 ±0.00004	0.00249 ±0.00002

OUT-OF-DOMAIN · HELD-OUT SITE FGMANAUS

Never touched during training

Method	PSNR ↑	SSIM ↑	SAM ↓	NDVI MAE ↓	M5 ↓
R0 Bicubic ×2	48.013	0.9583	0.837	0.01256	0.00074
R1 CNN, L1	44.087 ±0.133	0.9145 ±0.0025	1.564 ±0.042	0.02562 ±0.00048	0.00521 ±0.00012
R2 CNN, L1 + λ·SAM	43.956 ±0.098	0.9118 ±0.0023	1.568 ±0.026	0.02565 ±0.00083	0.00529 ±0.00009
R3 CNN, L1 + μ·consist	44.699 ±0.191	0.9256 ±0.0028	1.437 ±0.043	0.02345 ±0.00081	0.00448 ±0.00013
R4 CNN, Laplace NLL	43.719 ±0.139	0.9076 ±0.0026	1.594 ±0.042	0.02759 ±0.00139	0.00539 ±0.00013
R1a R1 + augmentation	45.857 ±0.055	0.9370 ±0.0007	1.159 ±0.005	0.02166 ±0.00045	0.00395 ±0.00004
R1m R1, multi-biome	47.067 ±0.083	0.9421 ±0.0005	0.967 ±0.024	0.01456 ±0.00034	0.00398 ±0.00009
R1ma R1, multi-biome + aug	47.609 ±0.037	0.9461 ±0.0004	0.861 ±0.010	0.01361 ±0.00024	0.00348 ±0.00001
R4m R4, multi-biome + aug	47.480 ±0.075	0.9461 ±0.0007	0.879 ±0.019	0.01432 ±0.00037	0.00357 ±0.00008

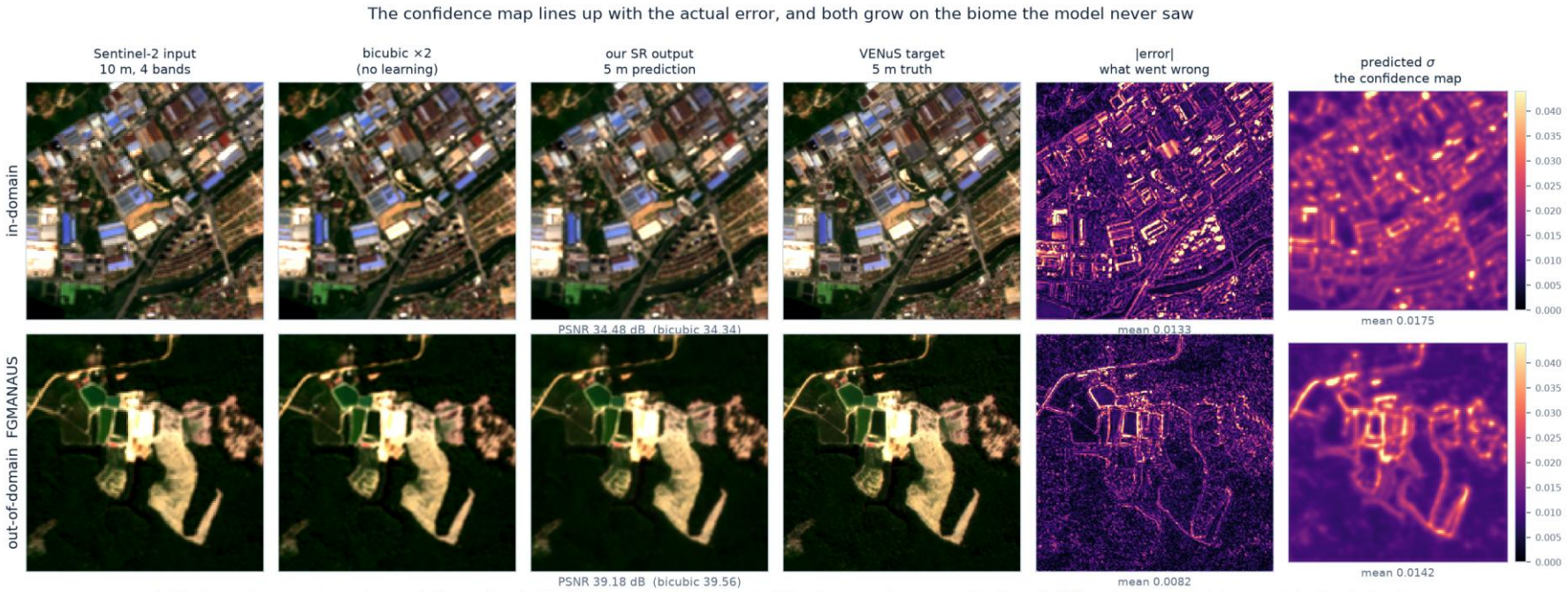
READING THE TABLE

Compare DOWN the out-of-domain table only. FGMANAUS is the same sealed 129 patches for every row. The in-domain split comes from the TRAINING sites, and the multi-biome rows train on six sites rather than two, so their in-domain column is a different, more varied set and is NOT comparable with the rows above it.

M5 rewards blur: bicubic invents no detail, so shrinking it back down reproduces the input and it wins that column by construction. Read M5 next to PSNR and SSIM, and note it depends on D (slide 7). A change smaller than the seed spread is not a change.

Results · what the confidence head actually produces

One real patch through the full pipeline, in-domain and on the unseen biome · the 2-minute demo



IS THE CONFIDENCE CALIBRATED, AND IS IT USEFUL?

Method / regime	AUSE ↓	AUSE/band ↓	ENCE ↓	Cv ↑	ρ ↑	PICP90
R4 head in-domain	0.174	0.353	0.0406	0.989	0.575	0.917
R4 head FGMANAUS	0.163	0.334	0.2835	1.236	0.570	0.875
R4m head in-domain	0.203	0.317	0.0906	0.846	0.530	0.919
R4m head FGMANAUS	0.119	0.452	0.2431	1.136	0.677	0.959
3-seed ensemble in-domain	0.329	0.506	n/a	n/a	0.389	n/a
3-seed ensemble FGMANAUS	0.275	0.352	n/a	n/a	0.445	n/a

WHAT EACH NUMBER IS FOR

- AUSE – rank pixels by σ, drop the worst; does error fall as fast as an oracle?
- AUSE/band – same, ranked INSIDE each band. Pooled AUSE flatters us (97% of top-quartile σ is NIR, simply 30× brighter). Quote this one.
- ENCE – is the stated spread the observed spread?
- Cv – does σ VARY? A constant map scores a perfect ENCE and is useless. Read the two together.
- PICP90 – does the 90% interval cover 90%? Laplace quantile, not a Gaussian z.

Why not the free 3-seed ensemble? It LOSES: AUROC 0.35, below chance, vs 0.76 for the head, and its disagreement does not rise on the unseen biome at all (×1.00). All three seeds learned the SAME prior, so they are identically wrong there. disagreement only sees what members disagree ABOUT.

Analysis · do the extra terms do anything?

A verbal objection turned into two measurements: gradient alignment, and the sensitivity of M5 to an operator nobody had justified

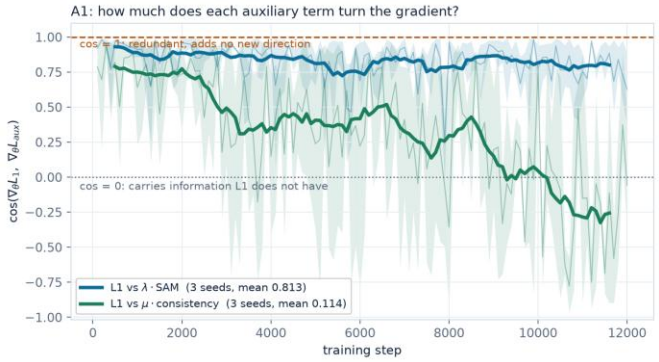
THE ARGUMENT WE HAD TO ANSWER

$$SAM(a \cdot p,\ t) = SAM(p,\ t) \quad \text{for every } a > 0$$

SAM is scale-invariant, so it only sees the DIRECTION of a pixel's 4-band vector. L1 constrains every band of every pixel, which forces the direction too. Both bottom out at the same parameter setting, so λ·SAM introduces no new optimum. That is settled analytically, not by experiment, and we concede it.

It leaves open whether the term still bends the optimisation PATH on the way there. That is measurable, and nobody had measured it.

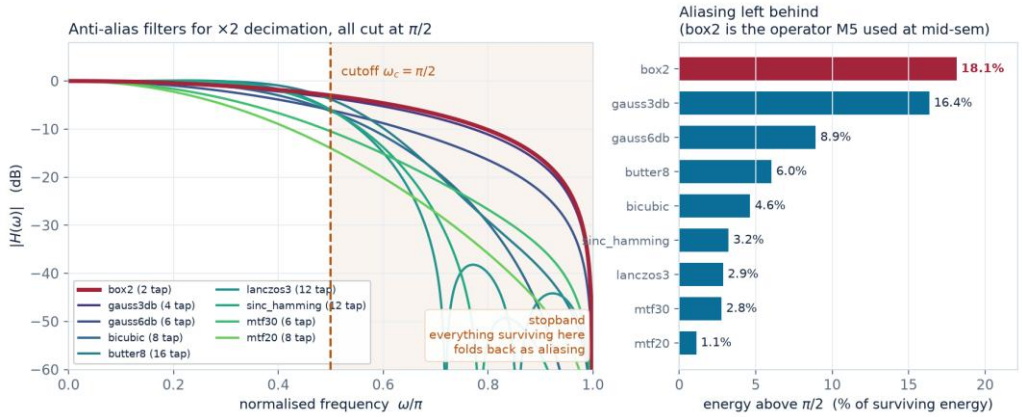
A1 · GRADIENT ALIGNMENT, AND WHAT IT PREDICTED



Term	Predicted	cos(∇L1, ∇aux)	OOD effect
λ·SAM	redundant: shares L1's optimum	0.81 ± 0.17	none, inside seed spread
μ·consist	independent: uses the INPUT	0.11 ± 0.59	better on all 5 metrics

SAM stays flat at 0.81 for all 12,000 steps. Consistency starts aligned, crosses zero near step 10,000 and ends NEGATIVE: late in training it pulls away from the target's biome-specific prior and toward the measurement, which holds on any biome, and that is what carries it across sites. (Yu et al., NeurIPS 2020.)

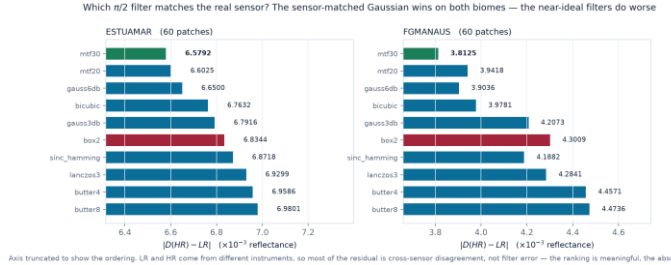
THE DECIMATION OPERATOR, AT π/2



How the filter choice reaches a reported number

Decimating by 2 keeps only frequencies below π/2; everything above folds back as aliasing. M5 is DEFINED through D, so the filter was silently shaping a column already on our results table. The 2x2 box is -3 dB at π/2 but only -8.3 dB at 0.75π, and leaves 18.1% of surviving energy in the stopband. A sensor-matched Gaussian leaves 1.1%.

WHICH FILTER MATCHES THE REAL INSTRUMENT?



Best on real pairs: ESTUAMAR mtf30 (4% better than box2) · FGMANAUS mtf30 (11% better than box2)

Analysis · generalisation and trust

The mid-sem failure, its diagnosis, what fixed it, and how the confidence head closes the gap to bicubic

THE FAILURE, AND THE ARITHMETIC BEHIND IT

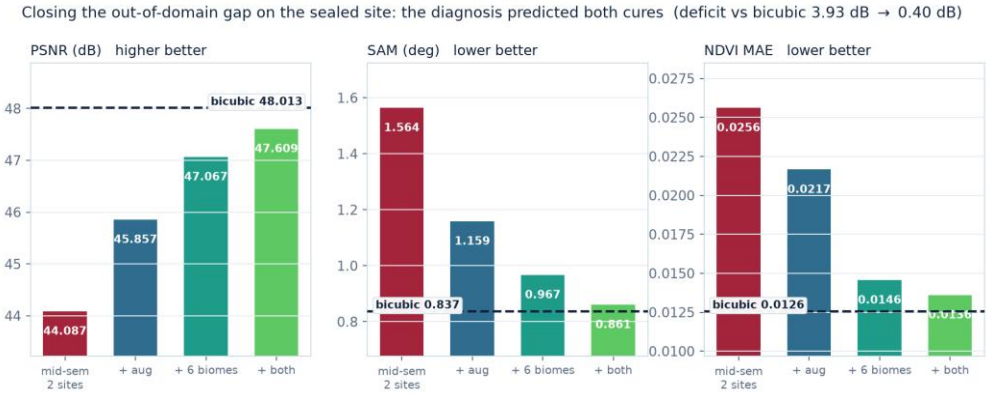
The model applies a brightness prior learned from European coast and farmland to Amazon rainforest.

For R1, the mid-sem model, mean signed error on the test site is +0.0025, +0.0050, +0.0030 on blue, green, red and +0.0001 on NIR: the three visible bands are biased UP and NIR is not. Against a true red of 0.004 a +0.003 bias is a 75% relative error.

true NDVI = (0.279 − 0.004)/(0.279 + 0.004) = 0.972 biased = (0.279 − 0.007)/(0.279 + 0.007) = 0.951

A shift of 0.021, which matches the measured NDVI degradation. The explanation is quantitatively consistent, not hand-waving. Bicubic has no learned prior, so it has no bias to carry across.

CLOSING THE GAP ON THE SEALED SITE



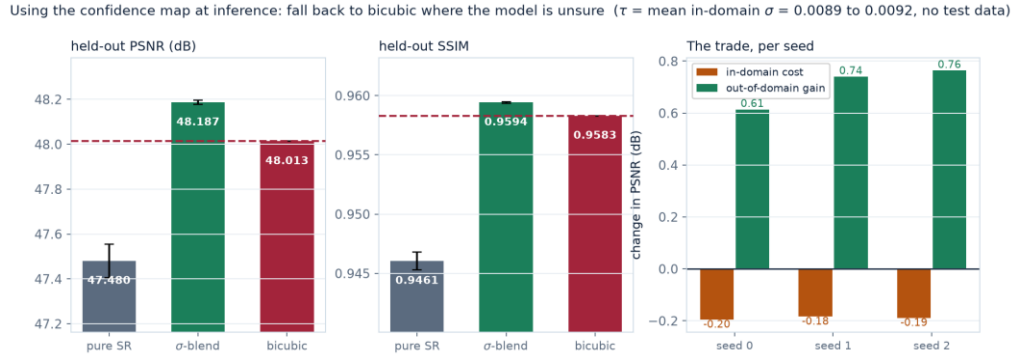
WHAT THE AUGMENTATION DID, MEASURED

A per-band gain applied identically to the LOW and HIGH resolution patch leaves the pair physically consistent while removing absolute brightness as something the model can lean on.

It works on the bands it was aimed at. Signed bias on FGMANAUS (×10⁻³): blue 2.50 → 1.34, green 5.00 → 2.27, red 3.00 → 2.04.

Concede the rest: NIR moves +0.10 → -2.84, so the visible-band prior is roughly halved rather than removed and some error is redistributed into NIR. Every out-of-domain metric still improves.

USING THE CONFIDENCE MAP, NOT JUST REPORTING IT



THE CONFIDENCE HEAD EARNS ITS PLACE

Blend the output toward bicubic where σ is high and the model beats bicubic on the held-out site: PSNR 48.187 ±0.009 against 48.013, SSIM 0.9594 against 0.9583. All three seeds win.

τ is not tuned on the test set. It is fixed to the model's own mean σ in-domain (0.0089 to 0.0092 across seeds), which reads as “trust the network where its uncertainty is typical of training”. Sweeping τ on validation fails instead, because in-domain the network already beats bicubic and so validation always prefers to blend less.

It is a trade, not a free lunch: -0.20 dB in-domain for +0.71 dB out of domain. SAM and NDVI improve over pure SR but stay behind bicubic.

The same σ also flags the shift unaided: AUROC 0.75.

CONCLUSIONS

- The confidence head is what closes the gap. Blending toward bicubic where σ is high beats bicubic out of domain on PSNR (48.187 vs 48.013) and SSIM (0.9594 vs 0.9583), with no retraining and no test-set tuning.
- The spectral loss term was redundant with L1. Conceded analytically, then measured: gradient alignment 0.81, and no metric moved outside the seed spread.
- The back-projection term compares the output with the INPUT, and it carries across sites: +0.61 dB PSNR and -8% NDVI error out of domain. Its gradient cosine sits near zero and swings widely, which is how a term carrying information L1 cannot see behaves.
- The mid-sem model failed across biomes because it learned an absolute brightness prior. Diagnosed to the band and to the NDVI arithmetic.
- Decimating at a π/2 cutoff changes a metric we had already reported: the 2×2 box leaves 18.1% of energy to alias, and the sensor-matched Gaussian models the real instrument best.